

$$\max_{\mathbf{x}} f(\mathbf{x}) = -1 - 7x_2 - 7x_3 + 5x_4 + 5x_5 + 8x_6 + 3x_7$$

where

$$x_1 = 7 - x_2 + 3x_3 - 4x_4 + 7x_5 + 6x_6 + 7x_7 \geq 0$$

$$x_8 = 3 + 6x_2 - 6x_3 + 2x_4 - 7x_5 + 2x_6 + 7x_7 \geq 0$$

$$x_9 = 5 - 3x_2 - x_3 - 2x_4 - 7x_5 - 7x_6 - 4x_7 \geq 0$$

$$x_2 = x_3 = x_4 = x_5 = x_6 = x_7 = 0$$

$$x_4 = 1.8, \quad x_1 = 0, \quad f(\mathbf{x}) = 7.8$$

		x_2	x_3	x_4	x_5	x_6	x_7
$f(\mathbf{x})$	-1	-7	-7	5	5	8	3
x_1	7	-1	3	-4	7	6	7
x_8	3	6	-6	2	-7	2	7
x_9	5	-3	-1	-2	-7	-7	-4